# Staying in Touch, Simply

## Co-Designing a Way for an Older Adult to Use His Smartphone (mid-term draft)

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Many older adults own a smartphone they cannot use. We are co-designing, with Manuel, an 80-year-old man in Lisbon, a way for him to use the smartphone his daughter bought him so he could follow the family's WhatsApp group. He does not want a simpler phone: he wants the smartphone's family features, photos, messages and news, without having to learn how to work it. This mid-term draft covers related work, secondary research on older adults in Portugal, our method, and the first two of three co-design sessions, in which Manuel framed the problem and chose a concept: a plain cover that shows one thing at a time, with a green button to talk and a red one to end.

CCS Concepts: • **Human-centered computing** → **Accessibility technologies**; *Participatory design*; *Mobile devices.*

Additional Key Words and Phrases: co-design, older adults, inclusive design, smartphones, voice interaction

## 1 INTRODUCTION

Smartphones are now where many families share news and photos, but many older adults find them hard to learn and easy to get wrong [14, 15]. In Portugal, which has the second-oldest population in the EU, only one in five people aged 65–74 has at least basic digital skills [10, 11].

Rather than designing for older adults, we design with one of them [25]. Over three sessions following the Double Diamond [6], Manuel defines the problem with us, creates and chooses ideas, and will test our prototype at home. This draft covers the first two sessions.

## 2 RELATED WORK

### 2.1 Older adults and smartphones

Older adults are not simply afraid of technology: they weigh its benefits against its hassle [18], and anxiety and confidence, more than age itself, predict whether they adopt it [4]. Touchscreens remain a barrier. Older adults' typing errors on touchscreens rise with hand tremor [20], and flat designs make it harder for them to tell what can be pressed [28]. Senior launchers make the screen simpler, but fewer features alone do not make a phone usable [21].

### 2.2 Voice interaction

Voice assistants can reach older adults who avoid screens, but they are abandoned when people find few valuable uses [27], reminders go unused when people doubt them [23], and recognition errors become the main frustration over time [13]. A voice interface should do a few things the person values and recover gracefully from mistakes.

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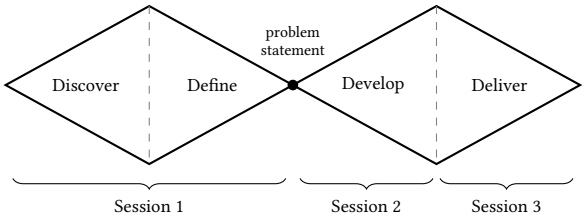



Fig. 1. Our three co-design sessions mapped onto the Double Diamond.

### 2.3 Co-design and stigma

Co-design treats the people who will use a design as co-creators [25], and participatory and generative methods help older people take part from the start [1, 17]. Designs must also avoid stigma: devices that mark their users as old or disabled are often rejected [19, 26].

## 3 METHODOLOGY

### 3.1 Co-designer

Manuel is 80, a widower and a retired electrician, and lives alone in Lisbon; his daughter introduced him to us. He calls his family every day on a button phone. In July, his daughter bought him a smartphone so he could join the family's WhatsApp group, where photos of his newborn great-grandson are shared. He does not know how to work it, so it stays in a drawer.

### 3.2 Sessions

We follow the Double Diamond (Figure 1). Each session takes about 70 minutes in Manuel's kitchen. In session 1 we used directed storytelling, a tour of both his phones and a communication map, and wrote the problem statement with him. In session 2 he sorted 16 action cards, compared three empty 3D-printed mock-ups, acted out the phone's behaviour while a teammate played its voice, and narrated a storyboard. Session 3 will test our prototype.

We adapted the sessions to him: he hosted a card game before any question, we read the consent form aloud in plain Portuguese, and only one of us asked at a time. With his consent, we took notes, recorded audio and photographed his hands and materials, never his face, and we analyse them with thematic analysis [3]. Participation follows a project declaration signed by Manuel and the team; recordings will be deleted by [date].

## 4 SECONDARY RESEARCH

Older adults in Portugal are less connected than their European peers (Table 1), but those who are online message and call more than the EU average, so staying in touch is a central use.

Existing products do not fit Manuel. Minimal phones such as the Light Phone drop apps like WhatsApp [16]. Simplified modes such

Table 1. Older adults in Portugal and the EU-27, Eurostat 2025 [7–11].

| Indicator | Portugal | EU-27 |
|---|---|---|
| Population aged 65 or over | 24.3% | 22.0% |
| Never used the internet, ages 65–74 | 36.1% | 16.4% |
| At least basic digital skills, ages 65–74 | 20.7% | 33.0% |
| Instant messaging, internet users aged 65–74 | 79.8% | 74.5% |
| People aged 65 or over living alone | 25.7% | 32.2% |

as Samsung's Easy mode only change how the phone looks, and Apple's Assistive Access leaves apps like WhatsApp in their usual design [2, 24]. Mainstream voice assistants do not support European Portuguese [12, 22].

## 5 EARLY FINDINGS

### 5.1 Session 1

- *He cannot work the smartphone.* His taps were too short, a pinch shrank the text and the keyboard opened by accident: "I broke the letters. Put them back."
- *But he wants what is inside it.* He will not refuse it, as his sister did, "because the refusal includes the baby".
- *The family's news is in the group.* He missed the new time of a blood test that his daughter wrote only in the group: "She wrote Pai in a conversation with twelve people, and she thought that was telling me."
- *He wants the name first.* After an "olá pai" scam text: "I still want to know the name before I see the child."

Quotes are our translations from Portuguese. Manuel corrected two drafts before signing this problem statement:

> Manuel needs a way to see the new photos of his great-grandson, to hear his great-granddaughter and the family group, and to reply with a short sentence, without learning the smartphone. He needs the sender's name first, because a message that says "olá pai" may be a scam.

### 5.2 Session 2

Manuel kept the cards about his family (Table 2) and wrote two of his own: the sender's name before any photo, and only people his daughter saved. Of three mock-ups, he rejected a bumper that left the icons visible and a separate box, "a third phone", and chose a full cover over his own smartphone that shows one thing at a time: "This. I see one thing." He asked for a green button to talk where his thumb rests and a red one to end.

In the role-play he set rules for the voice: no greeting, the sender's name first, silence while he looks, no photo from unknown senders and, after two misunderstandings, nothing at all: "Rude and safe. I prefer that to cheerful and wrong."

### 5.3 Lessons about the methods

Letting Manuel host a game before any question put him in charge, and options he could refuse made his choices clear. We were wrong that one button would be enough.

Table 2. Manuel's card sort in session 2. * Cards he wrote himself.

| Need | Would like | Don't need |
|---|---|---|
| Name before the photo* | Appointment reminders | Medication reminders |
| Only people his daughter saved* | Emergency help, only to call his daughter | Torch |
| Family photos | Video call, only the baby | Radio |
| Messages read aloud | Time and date | Take photos |
| Spot scam messages | | Weather |
| Calls, if as easy as his button phone | | Magnifier |
| Short spoken replies | | |

## 6 NEXT STEPS

Before session 3 we will build a demoable prototype: a dark 3D-printed cover for his smartphone with one window and green and red buttons, and an Android app that shows one photo at a time and reads messages aloud in European Portuguese. A teammate will steer the voice behind the scenes, as in a Wizard-of-Oz study [5]. Manuel will keep his button phone until calling his daughter on the new one is as easy as its green button.

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